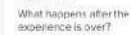
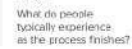
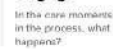
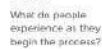


○ Clearly Larson	○ Morgan Wahner	○ Jennifer Phillips	○ Respond to Flores	○ Denise Sabo
				

Browsing, booking,
attending, and rating a
local city tour

How does someone initially become aware of this process?



Your support is the user profile Personalized recommendations	Personalized hour offers Personalized hour suggestions after new hour booking
The simplest way depends on the "user experience" it has to be a combination of all with their ability to obtain the greatest value	Personalized hour offers can deliver additional recommendations to customers already booked hourly reservations to increase their productivity
	The customer's schedule or what it says about their business (frequency of reservations) for their business
	Hourly reservation packages could be more effective, and it is a better way to increase productivity in their business



Help me see what I've done before

Help me see what I could be doing now

Help me see what I can do to enhance my new trip

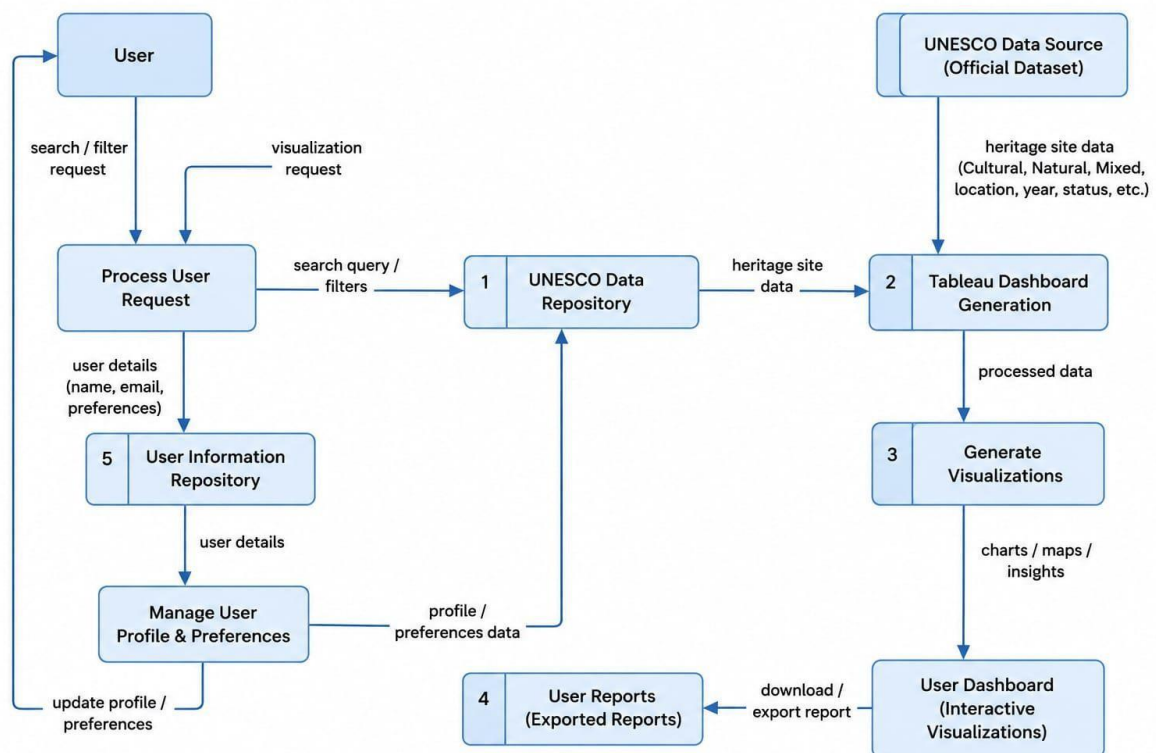


Project Design Phase-II

Data Flow Diagram & User Stories

Data Flow Diagram

A Data Flow Diagram (DFD) explains how data moves within an analytics system. Data is collected from different sources such as SQL databases, Excel, or CSV files. The data is cleaned, transformed, validated, and stored before visualization. Tableau connects to the processed data to build dashboards for business users. DFD consists of external entities, processes, data stores, and data flows. It simplifies system understanding, improves documentation, and helps developers and analysts communicate effectively.



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Dashboard User)	Access Dashboard	USN-1	As a user, I can open the Tableau dashboard to explore UNESCO World Heritage Sites.	Dashboard loads successfully.	High	Sprint-1
Customer (Dashboard User)	Dashboard Navigation	USN-2	As a user, I can navigate between multiple dashboards (Overview, Risk Analysis, Forecasting).	All dashboards open correctly.	High	Sprint-1
Customer (Dashboard User)	Region Filter	USN-3	As a user, I can filter heritage sites by region.	Selected region displays only relevant sites.	High	Sprint-1
Customer (Dashboard User)	Country Filter	USN-4	As a user, I can filter heritage sites by country.	Dashboard updates according to selected country.	High	Sprint-1
Customer (Dashboard User)	Heritage Category	USN-5	As a user, I can filter sites by Cultural, Natural, and Mixed heritage.	Correct category data is displayed.	High	Sprint-1
Customer (Dashboard User)	Map Visualization	USN-6	As a user, I can view UNESCO sites on an	Map displays site locations correctly.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			interactive world map.			
Customer (Dashboard User)	Trend Analysis	USN-7	As a user, I can view yearly growth of UNESCO World Heritage Sites.	Line chart displays yearly trends accurately.	High	Sprint-1
Customer (Dashboard User)	Danger Analysis	USN-8	As a user, I can identify heritage sites that are endangered.	Dashboard highlights endangered sites.	High	Sprint-1
Customer (Dashboard User)	Country Comparison	USN-9	As a user, I can compare the number of heritage sites between countries.	Comparison chart updates correctly.	Medium	Sprint-2
Customer (Dashboard User)	Forecasting	USN-10	As a user, I can view future forecasting of UNESCO heritage site growth.	Forecast chart predicts future trends.	Medium	Sprint-2
Customer (Dashboard User)	Dashboard Interaction	USN-11	As a user, I can click charts to filter other visualizations.	All charts interact dynamically.	High	Sprint-2
Customer (Dashboard User)	Tooltip Information	USN-12	As a user, I can hover over charts to view detailed information.	Tooltip displays complete details.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Dashboard User)	Export Dashboard	USN-13	As a user, I can download dashboard reports as PDF/Image.	Export completes successfully.	Medium	Sprint-3
Customer (Dashboard User)	Search Heritage Site	USN-14	As a user, I can search for a UNESCO heritage site by name.	Search returns matching site details.	High	Sprint-2
Customer (Dashboard User)	Dashboard Performance	USN-15	As a user, I want dashboards to load quickly.	Dashboard loads within acceptable time.	High	Sprint-3
User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Data Management	USN-16	As an administrator, I can update UNESCO dataset.	Updated data appears in dashboard.	High	Sprint-1
Administrator	Data Validation	USN-17	As an administrator, I can validate imported data before publishing.	Only valid data is displayed.	High	Sprint-1
Administrator	Dashboard Maintenance	USN-18	As an administrator, I can modify dashboard visualizations.	Changes reflect after publishing.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Report Generation	USN-19	As an administrator, I can generate summary reports.	Reports are downloadable.	Medium	Sprint-2
Administrator	User Access	USN-20	As an administrator, I can publish dashboards for users.	Users can access published dashboards.	High	Sprint-1

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Access	Open Tableau Dashboard, View Home Dashboard
FR-2	Data Visualization	Display Interactive Charts, Maps, Bar Charts, Pie Charts, Treemaps, Line Charts
FR-3	Filter & Search	Filter by Region, Country, Heritage Category (Cultural, Natural, Mixed), Year
FR-4	Interactive Dashboard	Cross-filtering, Highlight Actions, Interactive Tooltips
FR-5	UNESCO Site Analysis	Display Total Heritage Sites, Endangered Sites, Heritage Distribution by Region and Country
FR-6	Trend Analysis	Analyze Growth of UNESCO Heritage Sites Over Time
FR-7	Forecast Analysis	Predict Future Growth of Heritage Sites using Tableau Forecasting
FR-8	Geographic Analysis	Display UNESCO Sites on Interactive World Map
FR-9	Report Export	Download Dashboard as PDF/Image, Export Charts and Reports
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-10	Dashboard Navigation	Navigate Between Overview Dashboard, Heritage Analysis Dashboard, Danger Analysis Dashboard, Forecast Dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be simple, user-friendly, and easy to navigate with clear visualizations.
NFR-2	Performance	Dashboard should load quickly and filters should respond within a few seconds.
NFR-3	Reliability	Dashboard should display accurate UNESCO data without errors or missing information.
NFR-4	Availability	Dashboard should be accessible whenever users open Tableau Public or Tableau Desktop.
NFR-5	Scalability	Dashboard should support future UNESCO dataset updates and additional visualizations.
NFR-6	Maintainability	Dashboard should be easy to update with new UNESCO data and enhancements.
NFR-7	Compatibility	Dashboard should work properly on desktop browsers and Tableau Public.
NFR-8	Data Accuracy	Visualizations should accurately represent the UNESCO World Heritage dataset.
NFR-9	Security	Dataset should remain protected from unauthorized modification, and only authorized users should update the source data.
NFR No.	Non-Functional Requirement	Description
NFR-10	Accessibility	Charts, colors, labels, and tooltips should be readable and understandable for all users.

Project Design Phase-II

Technology Stack (Architecture & Stack)

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive dashboard for visualizing UNESCO World Heritage Sites	Tableau Desktop, Tableau Public
2	Application Logic-1	Data cleaning, preprocessing and transformation	Microsoft Excel, Tableau Prep (Optional)
3	Application Logic-2	Dashboard calculations, filters, parameters and calculated fields	Tableau Calculated Fields
4	Application Logic-3	Forecasting and trend analysis of heritage sites	Tableau Forecasting & Analytics
5	Database	UNESCO World Heritage Sites dataset	CSV Dataset (Kaggle/UNESCO Dataset)
6	Cloud Database	Dataset published with Tableau dashboard	Tableau Public Cloud
7	File Storage	Store dataset and Tableau workbook	Local File System (.csv, .twb, .twbx)
8	External API-1	UNESCO World Heritage information source	UNESCO Open Data / UNESCO Dataset
9	External API-2	Public dataset for visualization	Kaggle Dataset
S.No	Component	Description	Technology
10	Machine Learning Model	Time-series forecasting for future heritage trends	Tableau Forecast Model
11	Infrastructure (Server / Cloud)	Dashboard deployment and sharing	Local System & Tableau Public Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Dataset collected from open-source platforms and visualized in Tableau	Kaggle Dataset, Tableau Public
2	Security Implementations	Dashboard is published securely and source dataset is protected from unauthorized modifications	Tableau Public Permissions, Local File Security
3	Scalable Architecture	Dashboard can easily accommodate additional UNESCO records and new visualizations	Tableau Dashboard Architecture
4	Availability	Dashboard is accessible online through Tableau Public and locally using Tableau Desktop	Tableau Public Cloud
5	Performance	Optimized dashboard using filters, calculated fields, and efficient visualizations for fast loading	Tableau Performance Optimization